

IMPORT

```
import pandas as pd
```

MERGE HOW

```
left.merge(right, on='key', how='inner')
left.merge(right, on='key', how='left')
left.merge(right, on='key', how='right')
left.merge(right, on='key', how='outer')
```

INDEX / LABEL CHECKS

```
row_concat.iloc[3]
row_concat.loc[0]
df1.columns = ['A','B','C','D']
df1.index = [0,1,2,3]
```

CONCAT ROWS

```
pd.concat([df1, df2, df3])
pd.concat([df1, df2, df3], ignore_index=True)
pd.concat([df1, new_row_df])
pd.concat([df1, pd.DataFrame([data_dict])],
          ignore_index=True)
```

MANY-TO-MANY

```
visited.merge(survey, left_on='ident',
              right_on='taken')
site.merge(visited, left_on='name',
           right_on='site').merge(survey, left_on='ident',
                                right_on='taken')
```

CONCAT COLUMNS

```
pd.concat([df1, df2, df3], axis=1)
col_concat['A']
col_concat['new_col'] = ['n1','n2','n3','n4']
pd.concat([df1, df2], axis=1, ignore_index=True)
```

MANY-TO-ONE

```
site.merge(visited, left_on='name', right_on='site')
survey.merge(person, left_on='person',
             right_on='ident')
```

CONCAT LABELS

```
pd.concat([df1, df2, df3], join='outer')
pd.concat([df1, df2, df3], join='inner')
pd.concat([df1, df3], join='inner')
pd.concat([df1, df2, df3], axis=1)
pd.concat([df1, df3], axis=1, join='inner')
```

SUFFIXES

```
left.merge(right, on='key', suffixes=('_left',
                                     '_right'))
left.merge(right, left_on='a', right_on='b',
          suffixes=('_x','_y'))
```

MERGE BASIC

```
left.merge(right, on='shared_col')
left.merge(right, left_on='name', right_on='site')
```

ONE-TO-ONE

```
visited_subset = visited.loc[[0, 2, 6], :]
site.merge(visited_subset, left_on='name',
          right_on='site')
```

DO NOT USE

```
df.append(df2)
df1.append(data_dict, ignore_index=True)
```